11.7.5 Integration by parts

Integration by parts method is generally used to find the integral when the integrand is a product of two different types of functions or a single logarithmic function or a single inverse trigonometric function or a function which is not integrable directly. From the formula for derivative of product of two functions we obtain this useful method of integration.

If u and v are two differentiable functions then we have

$$d(uv) = vdu + udv$$

$$udv = d(uv) - vdu$$

Integrating

$$\int udv = \int d(uv) - \int vdu$$

$$\int udv = uv - \int vdu$$

$\int udv$ in terms of another integral $\int vdu$ and does not give a final expression for the integral $\int udv$. It only partially solves the problem of integrating the product $u\,dv$. Hence the term '**Partial Integration**' has been used in many European countries. The term "**Integration by Parts**" is used in many other countries as well as in our own.

The success of this method depends on the proper choice of u

(i) If integrand contains any non integrable functions directly from the formula, like $\log x$, $\tan^{-1} x$ etc., we have to take these non integrable functions as u and other as dv.

(ii) If the integrand contains both the integrable function, and one of these is x^n (where n is a positive integer) then take $u = x^n$.

(iii) For other cases the choice of u is ours.

Example 11.33

Evaluate the following integrals

(i) $\int xe^x dx$ (ii) $\int x\cos x\, dx$ (iii) $\int \log x\, dx$ (iv) $\int \sin^{-1} x\, dx$

Solution

(i) Let $I = \int xe^x dx.$

Since x is an algebraic function and e^x is an exponential function,

so take $u = x$ then $du = dx$

$$dv = e^x dx \Rightarrow v = e^x$$

Applying Integration by parts, we get

$$\int udv = uv - \int vdu$$

$$\Rightarrow \int xe^x dx = xe^x - \int e^x dx$$

That is, $\int xe^x dx = xe^x - e^x + c.$

(ii) Let $I = \int x\cos x\, dx$

Since x is an algebraic function and $\cos x$ is a trigonometric function,